DHEERAJ KUMAR

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EDUCATION

M.S, Robotics and Autonomous Systems **Graduating May 2024**
Arizona State University, Tempe, USA **3.96 GPA**
B.E, Production Engineering **2015 - 2019**
Jadavpur University, Kolkata, India **7.6 GPA**

TECHNICAL SKILLS

Programming: Python, MATLAB, R, C++, RAPID **Hardware:** Arduino, Raspberry Pi, Nvidia Jetson TX2
Software: Siemens NX, Minitab, AutoCAD, Gazebo, Tecnomatix Process Simulate, Robot Studio, ShopFloor Editor, CasADI, MuJoCo
Tools and Framework: ROS2, Tensorflow, OpenCV, Docker, Git, PyTorch, Tableau, MediaPipe

PROFESSIONAL WORK EXPERIENCE

Maruti Suzuki India Limited, Gurgaon, India: **Assistant Manager(Robotics Engineer)** **Jul 2019 - Jul 2022**

- Proficient in Online and Offline Robot programming for 6-axis painting robot and SGC(Stone Guard Coating) robot using teach pendant and Tecnomatix Process Simulate software.(***Models:- EPX2800, MPX3500 & IRB5500, Controller:- NX100, DX200 & IRC5P, Manufacturers:-Yaskawa & ABB, Type:- wall-mounted & floor-mounted 6-axis robot***)..
- Oversaw commissioning and installation of robots and robotic systems, jig design for robotic feasibility.
- Extensive experience in robotics mechanical and electrical troubleshooting, synchronization of conveyors and robots.
- Achieved the cost-saving of Rs 2.57 million/annum by optimizing robot teaching trajectories(changed the robot trajectories from along-the-conveyor to across-the-conveyor) and robot parameters.
- Working knowledge of I/O, robot calibration, interlocks, sensors, actuators, pneumatic system, SCADA system, PLC.
- Eliminated vibration (due to robot's motion) by 6% by minimizing jerk and adjusting robot's link position and orientation.
- Achieved the DFT(painting coat) values on BIW from 46% to 71% within the tolerance of ± 2µ from mean DFT value by implementing a new robot teaching trajectory and, optimizing pitch of robot teaching stroke & robot parameters.

OPEN SOURCE CONTRIBUTION

Ivy, London(Remote): **Machine Learning Research Engineer** **May 23 - Jun 23**

- Implemented the "cummax" function within PyTorch frontend, which was subsequently merged into their official codebase.
- Utilized Pytest to conduct thorough testing and validation of the implemented algorithm, using Pycharm IDE on Ubuntu.

ACADEMIC PROJECTS

Bike Counting System in Tempe City using Self-supervised Reidentification | Computer Vision, Deep Learning **Sept 23 - Present**
Research funded Project, Working in a team of two; Project-link

- Conducting research in bicycle counting using motion-sensitive cameras across 27 locations in Tempe city, with a dataset of 0.55 million street images, a cost-effective solution for the city transportation department.
- Developing computer vision models, including Swin Transformer, YOLOv8, DETR, Siamese networks, and EnlightenGAN, to enhance low-light images as a preprocessing step for precise bicycle classification, localization, and reidentification through object detection and contrastive learning.

ML based Automated Disease Classification and Color Extraction by drone | Python, ML, ROS2, Docker **Feb 23 - May 23**
Coursework Project, Worked Individually; Project-link

- Trained the model with 2k plant leaves for three different classes based on pre-trained Efficient architecture in PyTorch frame.
- Integrated Tello drone with machine learning models for image classification and color extraction as publisher and subscriber nodes, respectively, using ROS2 framework in a Docker container for efficient deployment and management.

CNN-Based Automated Sorting System for Effective Recycling | Python, Computer vision, ML, ROS2 **Feb 23 - May 23**
Coursework Project, Worked in a team of four; Project-link

- Sorted recyclable and non-recyclable waste by MyCobot-M280 robotic arm using EfficientNet by leveraging transfer learning for real-time object detection in Tensorflow framework, annotated 500 trash objects for 11 classes for training our model.
- Performed the integration of gripper, robot control and Object detection ML model under the ROS2 framework.

Comparative Analysis of Machine Learning Models | Python, Machine Learning **Feb 23 - May 23**
Coursework Project, Worked in a team of two; Project Link

- Performed extensive analysis on used car dataset, comparing regression models like linear regression, Lasso, Ridge, decision tree, random forest, gradient boosting, and neural networks, and evaluating key metrics.
- Neural network model achieved superior performance but required extensive hyperparameter tuning and computational resources. In contrast, linear regression performed well with limited resources, minimizing tuning needs.

ACADEMIC WORK EXPERIENCE

Intelligent Robotics and Interactive Systems Lab, ASU **Research Assistant** **Jun 22 – Present**

- Learnable geometry for collision detection and robot manipulation. This involves the use of implicit functions to represent 3D shapes, followed by the application of optimization-based collision detection algorithms for accurate collision analysis, along with employing Model Predictive Control (MPC) techniques for manipulation.

AWARD & CERTIFICATE

- Applications of AI for Anomaly Detection(using XGBoost, Autoencoders, GANs), NVIDIA DLI. link **Nov 23**
- Accolade with reward and recognition title for quality improvement with the help of painting robots, Maruti Suzuki. **Mar 21**